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FUNDAMENTAL PRINCIPLES OF DATA SCIENCE MASTER'S
THESIS

Predicting venous thromboembolism in cancer patients using contrastive learning

Author:

Júlia VILLARÓ CAÑIZAL

Supervisors:

Dr. Oriol PUJOL VILA
Bárbara LOBATO DELGADO

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Abstract

Facultat de Matemàtiques i Informàtica

MSc Fundamental Principles of Data Science

Predicting venous thromboembolism in cancer patients using contrastive learning

by Júlia VILLARÓ CAÑIZAL

This thesis studies whether contrastive learning can improve the prediction of venous thromboembolism (VTE) in cancer patients using a particular dataset, which contains very few samples and a large number of clinical variables. In this setting, standard supervised learning models often struggle to capture the complex patterns required for accurate prediction. To address these limitations, the work investigates whether contrastive learning can learn more robust patient representations from limited data.

It's also examined whether adding a feature gating mechanism to the contrastive model can down-weight less informative variables, which would improve model interpretability.

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Chapter 1

Introduction

1.1 Motivation

Venous thromboembolism (VTE) is a serious and common complication in patients with cancer, representing the second cause of death among them. Still, it remains difficult to predict it using current predictive clinical models, which are often based on limited variables and show little predictive performance.

The ONCOTHROMB project was created to support healthcare professionals by addressing this gap. It's goal is to develop data-driven tools to improve the diagnosis, prevention, and treatment of cancer-associated thrombosis. This project works with a smaller data subset called ONCOTH2_0.2.3, which doesn't contain genomic variables.

In such settings, standard supervised learning methods often struggle to learn the complex, non-linear patterns associated with VTE risk. This motivates the use of more advanced techniques to learn representations. Contrastive learning offers a promising alternative by enabling models to learn robust patient embeddings even when data are scarce.

Additionally, incorporating a feature gating mechanism offers a practical way to reduce the dimensionality of the dataset and highlight the most informative variables. This becomes especially useful in settings with many features, where selecting the most relevant variables is essential for building reliable models.

These factors suggest that combining contrastive learning with gating mechanisms could help improve ONCOTHROMB_2's models.

1.2 Goal

The primary objective of this thesis is to evaluate whether contrastive learning can improve the prediction of venous thromboembolism in cancer patients using the reduced ONCOTH2_0.2.3 dataset, which contains very few samples and many clinical variables. The central goal is to determine whether contrastive learning outperforms standard Neural Networks when data are limited.

A second objective is to develop a feature gating mechanism capable of identifying the less and most informative variables and reducing the dimensionality of the dataset. This component aims to improve interpretability in scenarios where the number of features is large relative to the sample size.

1.3 Contributions

The outcomes of this thesis will contribute to the `ONCOTHROMB_2` project by investigating whether contrastive learning and feature gating mechanisms can enhance representation learning and predictive performance. In addition, improving VTE risk classification can benefit scientific research, clinical decision-making, and healthcare outcomes, ultimately improving the quality of life of oncology patients and supporting changes in clinical practice with meaningful impact on the Spanish National Health System.

1.4 Structure

This thesis is structured to address the two central research questions:

1. Does contrastive learning improve the predictive performance in small, high-dimensional clinical datasets?
2. Can we incorporate a feature gating mechanism to help identify the most informative variables?

Chapter 3 presents the experimental setup required to reproduce the work. It performs an exploratory data analysis of the `ONCOTH2_0.2.3` dataset, and it describes the preprocessing pipeline (including cleaning, feature engineering, and handling missing values).

This chapter also introduces the machine learning models and describes the most important architectures considered in the main experiments: three standard supervised models (Random Forest, Logistic Regression and Gradient Boosting), a Neural Network embedding model combined with a Random Forest classifier (whose predictive results will later be compared against the contrastive ones), and the contrastive model. The feature gating added to the contrastive model is also exposed.

Chapter 4 contains the experimental results structured around the two research questions, and reports the performance of the models according to the evaluation metrics used throughout the study. It provides the central comparison between the accuracy on the predictions between the supervised baseline introduced above and the contrastive model.

Finally, the role of feature gating is examined to determine its contribution to variable selection and interpretability. This is done by adding a feature gating mechanism to the previous contrastive learning architecture. Its goal is to introduce a learnable weight to each input feature and push less informative variables to smaller weights.

The thesis concludes with a discussion of the findings and a summary of the main conclusions. Additional material is provided in the appendices: Appendix A details the preprocessing steps, Appendix B reports supplementary experimental results and exposes how the synthetic dataset for reproducibility was built, and Appendix C contains the GitHub repository with all notebooks and source code.

Chapter 2

Background and state of the art

Contrastive learning is a representation learning technique that trains models to distinguish similar from dissimilar observations by pulling positive pairs together in an embedding space while pushing negative ones apart. Basically, instead of relying only on the few available labels, these models learn by looking at the relationships within the data itself.

This "learning by comparing" approach is especially useful for small datasets that contain a large number of variables, because it encourages the model to extract invariant and robust features that generalize better when labels are limited (Le-Khac, Healy, and Smeaton, 2020). Consequently, contrastive learning can improve predictive performance beyond what is achievable with traditional, label-dependent architectures.

The **Khorana Risk score** is a widely used clinical tool designed to identify cancer patients starting chemotherapy who are at high risk of developing venous thromboembolism (VTE). It categorizes patients into low, intermediate, or high-risk groups to help determine if preventative blood-thinning medication (thromboprophylaxis) is necessary.

A relevant study was conducted in 2018 (Martín et al., 2018), in which they developed the **TiC-Onco score**, a clinical-genetic model for predicting VTE in cancer patients, combining clinical variables with genetic variants previously associated with thrombosis risk. The resulting model was compared with the Khorana score and showed a better predictive performance, achieving an AUC of 0.73 compared with 0.58 for the Khorana score. Furthermore, the model identified a larger proportion of patients who eventually developed VTE while maintaining a similar specificity. These results highlighted the potential value of incorporating genetic information into VTE risk assessment.

Building on this work, there was another development in 2022 (Muñoz et al., 2022): the **ONCOTHROMB score**. By combining clinical data with a large set of genetic markers, this model improved predictive accuracy, achieving an AUC of 0.781 (development) and 0.686 (external validation). It outperformed the Khorana score and improved the reclassification of patients previously misidentified as low risk. These findings confirmed the value of combining clinical and genetic data, establishing the ONCOTHROMB project as a benchmark for advanced predictive modeling.

ONCOTHROMB_1 and ONCOTHROMB_2 are two observational, prospective non interventional, and multicenter cohort studies conducted across different hospitals in Spain, designed to investigate VTE risk in cancer patients through the analysis of clinical, demographic, and genetic data.

The original cohort, ONCOTHROMB_1, has served as the basis for the development of multiple VTE prediction models. More recently, ONCOTHROMB_2 was initiated as a

separate cohort that includes a larger number of patients and a broader set of variables, due to the incorporation of newly identified risk factors and increased research resources. However, at the time of writing this thesis, there are no published predictive studies based on ONCOTHROMB_2.

Consequently, the current state of the art referred to the ONCOTHROMB framework is represented by the studies conducted on ONCOTHROMB_1, which constitutes the direct predecessor of the dataset used in this thesis.

Importantly, the goal of this thesis is not to develop a new clinical score nor to improve the Khorana, TiC-Onco, or ONCOTHROMB scores or models. Instead, the objective is to investigate whether contrastive learning and feature gating mechanisms can enhance representation learning and predictive performance when applied to the ONCOTHROMB2_0.2.3 dataset, which doesn't contain genomic variables.

Chapter 3

Experimental setup

To comply with the confidentiality and data-protection of the ONCOTHROMB_2 project and database, the original ONCOTH2_0.2.3 dataset can't be publicly released. The repository therefore includes a synthetic replica that preserves the statistical structure of the data while ensuring that no patient-identifying information is disclosed. Basically, it's provided only for reproducibility in the accompanying notebooks. However, all experimental results reported are obtained using the real dataset.

The complete implementation, along with the synthetic dataset used for reproducibility, is provided in Appendix C, which links to the project's GitHub repository.

3.1 Data preprocessing

The main dataset ONCOTH2_0.2.3 contains clinical and medical information for a total of 432 patients and 290 different features. It includes a wide range of variables, such as personal characteristics (age, weight, sex), clinical history (previous VTE, cancer type), and blood test results (d-dimer, hb, etc.). It's the core dataset used for modelling. Its data preparation process was extensive; therefore, Appendix A provides a more detailed description of the full pipeline.

3.1.1 Getting started

The first stage of the project focused on understanding the ONCOTH2_0.2.3 dataset and providing an initial overview of the available variables and their suitability for the prediction of venous thromboembolism (VTE).

First, patients who hadn't given informed consent to the study or who didn't meet the study's clinical eligibility criteria were excluded, based on the corresponding indicator columns in the dataset. This filter removed 37 patients, leaving a final cohort of 395.

Given the clinical relevance of age, an explicit age variable was created. When a patient had a recorded date of death, it was computed relative to that date; otherwise, age was calculated with respect to a fixed reference point (01-01-2026).

Most patients were Caucasian, followed by smaller groups of Latin-American, African, and other backgrounds. The sex distribution comprised 233 men (59%) and 162 women (41%). Data were collected across ten Spanish hospitals, reflecting the multicentric nature of the ONCOTHROMB_2 project.

Patient Inclusion and Blood Variables

An important methodological was taken regarding the clinical validity of the blood variables to predict VTE, because they come from a blood test that could be

taken before, after, or on the same day as the recorded VTE. This is why two strategies were considered:

- Keep only the patients whose blood tests were taken before the VTE, and therefore including blood features to the prediction model.
- Prioritizing sample size by keeping all patients and discarding blood test variables.

Given the limited number of available samples and the clinical ambiguity of blood measurements taken shortly before a thrombotic event¹, the second option was selected, leaving a total of 395 patients and 301 columns.

3.1.2 Feature selection and data cleaning

Because the dataset contained 301 variables but only 395 patients, it was necessary to reduce and clean the clinical variables to remove redundancy and low-signal information.

The process began by restricting the dataset to a set of clinically relevant variables provided by an expert familiar with both the dataset and its medical context. This reduced the number of features to 80. Still, several pairs of columns encoded the same information in different formats, so the version with greater granularity or numerical structure was retained.

A frequency-based score quantified how differently each variable behaved across VTE groups, helping identify features more strongly associated with VTE. Furthermore, a mixed-type correlation matrix was then constructed to examine dependencies between variables of different types (numeric and categorical). Thanks to this matrix, I was able to identify redundant features and understanding the dataset's internal structure, and as a consequence several variables were removed.

Finally, categorical variables were encoded numerically, and a second correlation matrix confirmed that the remaining features showed no concerning levels of correlation. The resulting cleaned dataset, containing 64 features, was then ready for the imputation of its missing values.

3.1.3 Final preprocessing pipeline

Building on the cleaned dataset, the next step consists of imputing all missing values to produce a fully usable modelling dataset. This stage combines simple and imputation strategies, guided by the previous correlation matrix.

Before performing any imputation, four rows were removed because they corresponded to patients with missing VTE status, and since it is the target variable, these cases can't be meaningfully imputed. Also, the variables `VTE_recurrence` and `VTE_concomitant_cancer` were discarded, since they are only defined when VTE is present.

For the imputation of the missing values, different strategies were used. If a variable showed no meaningful correlation with others, its missing values were imputed by sampling from its own empirical distribution. When two variables were correlated, their missing values were imputed jointly, using conditional and joint probability distributions to ensure that the completed values remained consistent with each other.

¹Blood test values obtained shortly before a VTE episode may already reflect physiological alterations caused by the developing thrombosis, making them unreliable as predictive features.

After completing all imputations, the final dataset contains 391 patients, of whom 95 experienced VTE (24%) and 296 did not (76%); the cohort includes 162 women (41%) and 229 men (59%). This fully imputed dataset is the one used as input for all the predictive models.

3.2 Supervised models

With the fully imputed dataset, the next step is to benchmark a set of supervised learning models. These models serve as baselines for understanding the difficulty of the task and for comparing the performance of the contrastive learning approach. Given the sample size and the class imbalance, they provide a first look at how conventional algorithms handle the structure and complexity of the data.

Random Forest classifier

As a first baseline, a Random Forest classifier was trained directly on the imputed clinical dataset. The goal of this model is to offer a reference for how a standard supervised method handles the structure of the data.

The pipeline is simple: the target variable (VTE) is separated from the predictors, and the dataset is split into training and test sets using a stratified split to preserve the class imbalance. A Random Forest with 101 trees and default depth was then fitted on the training data. Because VTE is the minority class, the model uses `class_weight="balanced"` so that misclassifying VTE cases is penalized more heavily during training.

Logistic regression

A Logistic Regression model was also used as a baseline to assess how a linear classifier performs on the clinical dataset. As with the Random Forest classifier, a stratified train–test split was applied to preserve the class imbalance, and the model was trained using the `lbfgs` solver because it provides stable convergence for high-dimensional logistic regression and supports class-weighting. The classifier was also fitted with `class_weight="balanced"` because of the class imbalance, giving additional weight to positive samples during training.

Gradient boosting

Gradient Boosting was included as another baseline model to evaluate how an ensemble of weak learners performs on the clinical dataset. Like the previous approaches, a stratified train–test split was applied to preserve the class imbalance. A standard `GradientBoostingClassifier` with default hyperparameters was trained on the resulting training set.

This model builds trees sequentially, with each new tree correcting the residual errors of the previous ones, allowing it to capture patterns without extensive tuning.

Embedding Network with Random Forest Classifier

This model combines a Neural Network for supervised representation learning with a Random Forest classifier trained on the learned representations. Its objectives are both classifying patients with respect to VTE and learning a latent representation of the original tabular data.

The architecture serves as the baseline of this work. The same encoder architecture will later be used in the contrastive learning approach, which allows a comparison between both models. The complete pipeline consists of two sequential stages:

1. **Representation learning:** a Neural Network is trained using the VTE labels to generate an embedding for each patient.
2. **Classification:** a Random Forest classifier is trained on the learned embeddings instead of the original features.

First, the **dataset** is divided into training and testing using a stratified train-test split. After splitting, the NumPy arrays are converted into PyTorch datasets through the `TabularDataset` class so that the data can be handled by PyTorch and fed to the Neural Network in mini-batches. Then, the **encoder** generates the embeddings so that the model is able to distinguishing between VTE and non-VTE patients. It contains two stages:

1. **Hidden layer:** The first layer projects the original feature space (of dimension $X.shape[1]=39$) into a 32-dimensional hidden representation. Layers are normalized and a LeakyReLU activation function introduces non-linearity.
2. **Embedding Layer:** The embedding dimension is fixed at $z \in \mathbb{R}^{16}$. Therefore, each patient is represented by a compact 16-dimensional vector. These embeddings constitute the latent representation learned by the network.

However, the encoder can't learn meaningful representations without a training objective. To provide a supervised learning signal, a temporary linear classification head is added. During training, the logit is compared with the true VTE label using Binary Cross Entropy with Logits Loss. By forcing the model to predict VTE status, the encoder gradually learns embeddings that separate the two classes. Once learned, the classification head is discarded and only the encoder is retained.

Next, **training** is performed in mini-batches of size 16. Each batch has its gradients reset and a forward pass through the network, which produces logits and embeddings. The classification loss is then computed and the model parameters are updated using the AdamW optimizer. Because of class imbalance, a positive-class weight is added into the loss function, increasing the penalty for misclassifying VTE cases and encouraging the model to focus on the minority class. The model is trained for 50 epochs with a learning rate of 10^{-4} and weight decay² of 10^{-4} .

The final loss is computed as the average over all samples. After training, the trained encoder is used to **transform** each patient into a 16-dimensional embedding, which is then used as input to the Random Forest classifier. Finally, the last **classification** step is done using a Random Forest classifier, which receives the embeddings. Its architecture is the same as in the baseline.

3.3 Contrastive model

The training objective of a contrastive model is to learn representations by comparing pairs of samples. The main idea is to pull positive pairs (different patients belonging to the same VTE class) closer in the embedding space and push negative pairs farther apart. Specifically, the model used consists of:

²batch_size and weight_decay were fixed to 16 and 10^{-4} after testing some combinations (see Appendix B).

- Generating supervised **positive and negative pairs** from the labelled dataset.
- Training an **encoder** (Neural Network) with a contrastive loss to structure patients in a latent space.
- Applying a **Random Forest classifier** on 16-dimensional embeddings for prediction.

This model is **supervised** because the VTE labels are used to build positive and negative pairs. It will also follow the same overall architecture as the supervised NN+RF³ baseline, to ensure a fair comparison between both methods. Consequently, any differences in predictive performance can be attributed to the representation-learning strategy, and this section only describes these methodological differences.

Some **hyperparameters** differ from those used in the previous model: the batch size was set to 32, the learning rate was increased to 5×10^{-4} , the weight decay was reduced to 10^{-8} , which also made the number of epochs increase until 100. These values provided faster and more stable convergence. This is justified because contrastive learning follows different optimization dynamics, and therefore it may require different hyperparameters to reach stable convergence. Changing these parameters don't avoid a fair comparison between the two models.

First, because contrastive learning requires pairs of samples, for each patient, a second patient is sampled dynamically during training. The proportion of VTE in the dataset is 24% (95 patients), but even though the probability for two samples of belonging in the same class is $\frac{\binom{95}{2} + \binom{296}{2}}{\binom{391}{2}} \sim 0.63$, the contrastive model uses `pos_fraction=0.5`, which is meant to control the proportion of positive and negative pairs during training and prevent the model from being dominated by the majority class.

A main architectural difference with respect to the previous model is in the network output. Now, two samples are passed through the same **encoder**, producing two embeddings, and the similarity between them is computed through their dot product. A high similarity means that the two patients are close in the latent space.

Another difference lies in the **loss function**: it corresponds to a hinge-based contrastive loss, which penalizes positive pairs when their similarity is too low and vice versa. This implies that the model is building an embedding space in which patients sharing the same class are grouped together, while patients belonging to different classes are separated. A **margin** parameter⁴ of 0.3 was introduced into the loss function, which means that if the similarity falls below that margin, no more penalty is applied.

3.3.1 Feature gating

This model adds a feature gating mechanism to the previous contrastive learning architecture, so most of the pipeline is the same. The only changes are the introduction of a learnable gating layer at the input of the encoder and an $L1$ penalty in the loss, which regularizes the feature weights.

³Neural Network for the embeddings, and Random Forest to classify.

⁴This margin was selected after evaluating different values (see Appendix B).

Because of the new gating system, the learning rate was increased from 5×10^{-4} to 10^{-3} . This and $\lambda_{L1} = 0.1$ were chosen to assure the best prediction performance of the model.⁵ However, the main part of this model is the **gating layer**, which gives a learnable weight to each input feature, all initialized to 1 so that every variable contributes equally at the start. Then, in the forward pass, each feature is multiplied by its gating weight and then fed into the encoder, whose architecture remains identical to the previous contrastive model.

The contrastive learning objective remains unchanged (the embeddings production, the samples' similarity and the hinge loss), and the gating layer does not alter the contrastive objective itself; instead, it adds an $L1$ penalty to the **loss** to encourage the gating weights to become sparse and pushes uninformative features to smaller values, while allowing informative ones to retain larger weights. Basically, at the same time, the model learns which are the most and less important features, and it learns the embedding space.

The training procedure is also the same as in the contrastive model. The only new step is when doing the loss computation, because now the $L1$ penalty is added to the objective function.

Even though the weight decay remained at 10^{-8} and the margin at 0.3, some **hyperparameters** differ from those used in the previous contrastive model:

- The learning rate was increased to 10^{-3} , because now we have more trainable parameters that must be optimized together with the encoder.
- The number of epochs increased to 100 so the model could learn the latent representation and the feature weights.

After training, embeddings are extracted like in the contrastive model, and each patient produces a 16-dimensional embedding vector through the encoder. Then, same as before, the embeddings are used to train the Random Forest classifier, whose settings also remain unchanged.

The main advantage of this gating model is that the final gating weights provide an estimate of each feature's performance during representation learning, offering a numerical way to measure their importance. The results and evaluation of this model are presented in Chapter 4.

⁵Combinations of learning rate and λ_{L1} were evaluated to identify the best values for prediction (see Appendix B).

Chapter 4

Experimental results

4.1 Evaluation metrics

Because the dataset is clinical and contains class imbalance, accuracy alone is not a sufficient metric. This is why performance was evaluated through:

- **Discrimination Capacity:** Measured via the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC).
- **Precision-Recall Trade-off:** Measured via the Precision-Recall (PR) curve and the Average Precision (AP), which provides a more rigorous assessment of the model's ability to identify the minority class (VTE).

Each model was evaluated across several train–test splits, allowing us to examine how their discriminative ability changed when they had less training data available. For every test size, the reported ROC-AUC, PR-AUC correspond to the average performance over multiple runs with different random states. A detailed description of the experimental settings and additional results is provided in Appendix B.

4.2 Supervised models

The three basic supervised models (Random Forest, Logistic Regression and Gradient Boosting) were evaluated to establish a simple baseline and obtain an initial view of how well standard methods predict VTE. Among them, the Random Forest achieved the highest accuracy by a clear margin (see Figures 4.1 and 4.2). However, relying solely on a Random Forest is not appropriate in this clinical context, because predicting VTE in cancer patients requires models that capture high-dimensional patterns and provide stable representations.

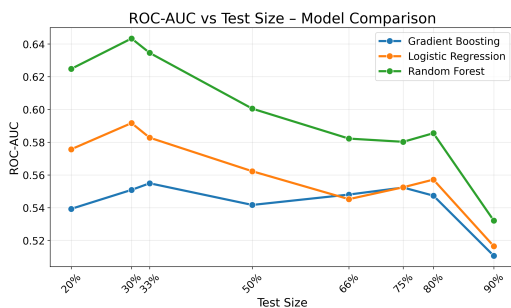


FIGURE 4.1: ROC-AUC comparison for the baseline models

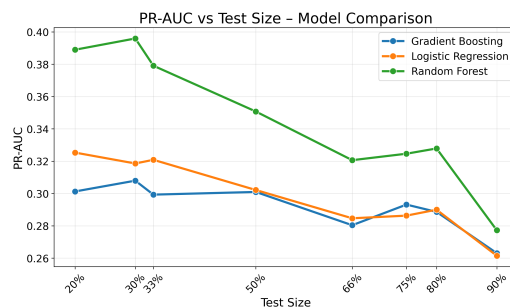


FIGURE 4.2: PR-AUC comparison for the baseline models

Still, these results were informative, as they indicated which classifier was most effective at separating the classes. For this reason, the Random Forest was selected as the classifier for the embedding-based models.

4.3 Neural Network

The embedding Network with Random Forest classifier model was evaluated across different test set sizes to analyze how its predictive ability changed when trained with varying amounts of data.

Test size	ROC-AUC	PR-AUC
20%	0.5791	0.3529
33%	0.5594	0.3296
50%	0.5363	0.2971
66%	0.5210	0.2756
75%	0.5297	0.2812
90%	0.4997	0.2494

TABLE 4.1: ROC-AUC and PR-AUC across different test sizes, for an Embedding Network with Random Forest classifier

The results clearly get worse in performance as the test size increases and the amount of training data decreases. Both ROC-AUC and PR-AUC drop when the model is trained on smaller subsets of the dataset, indicating that this Neural Network struggles to generalize in when data is scarce. Overall, the model shows limited robustness when trained with reduced information.

4.4 Contrastive learning

This experiment was essential for evaluating whether supervised contrastive learning improves predictive performance when the amount of available training data is limited. The model was evaluated across different test set sizes.

Test size	ROC-AUC	PR-AUC
20%	0.5687	0.2976
33%	0.5711	0.2916
50%	0.5896	0.3161
66%	0.5584	0.2881
75%	0.5518	0.2801
90%	0.5472	0.2757

TABLE 4.2: ROC-AUC and PR-AUC across different test sizes, for the Contrastive model

The results show that, as in the previous model, both ROC-AUC and PR-AUC decrease when the test size increases and the amount of training data becomes smaller. However, the contrastive model achieves higher performance than the non-contrastive Neural Network across the majority of test sizes, and its decline in performance is much slower (see Figures 4.3 and 4.4).

These findings allow us to answer the first research question: *Does contrastive learning improve predictive performance in small, high-dimensional clinical datasets?* The

results indicate that it does, so contrastive learning does provides a more robust representation that leads to better predictive performance, even when the amount of training data is limited.

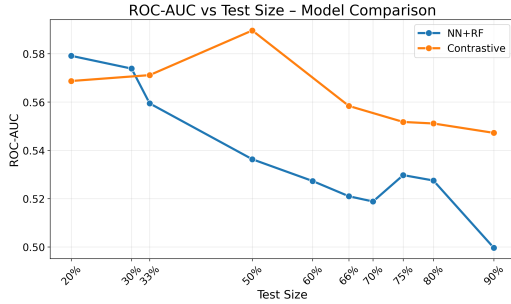


FIGURE 4.3: ROC-AUC comparison

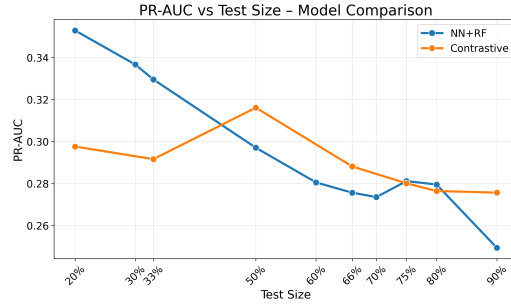


FIGURE 4.4: PR-AUC comparison

4.4.1 Feature gating

To address the second research question (*Can we incorporate a feature gating mechanism to help identify the most informative variables?*), a gating layer was added to the contrastive model. Even though with the model would push some gating weights close to zero, the final weights were not as small as expected. Still, there was a clear difference between the biggest and smallest weights (see Appendix B).

To evaluate whether the gating mechanism was genuinely identifying meaningful variables rather than producing unstable or random patterns, a robustness analysis was performed across three regularization strengths: $\lambda_{L1} \in \{0.05, 0.1, 0.2\}$. The resulting average gating vectors were compared using two strategies:

- Rank–rank scatterplots (Figure 4.5) showed that the rankings remained mostly consistent across all three settings, with points closely aligned along the diagonal.
- Spearman rank correlations were computed between the three ranking vectors. As shown in the correlation heatmap (Figure 4.6), all off-diagonal correlations exceeded 0.83, confirming strong monotonic agreement across the different λ_{L1} values.

The scatterplots show that the model is most consistent in identifying the least important features, which cluster along the diagonal across all combinations of λ_{L1} values. This confirms that the gating mechanism is stable and not producing random weight patterns. Overall, the gating layer does well on picking out the variables that matter less.

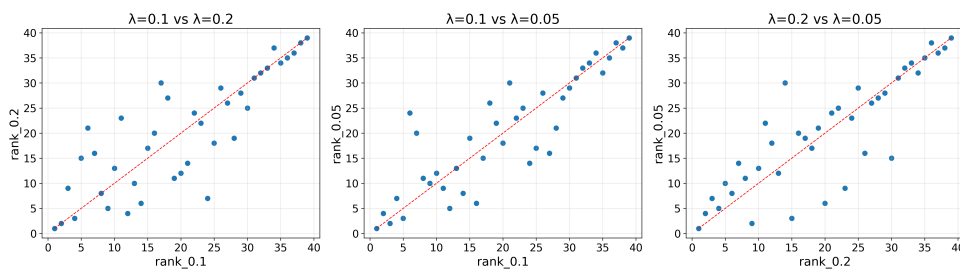


FIGURE 4.5: Scatterplots comparing feature-importance rankings

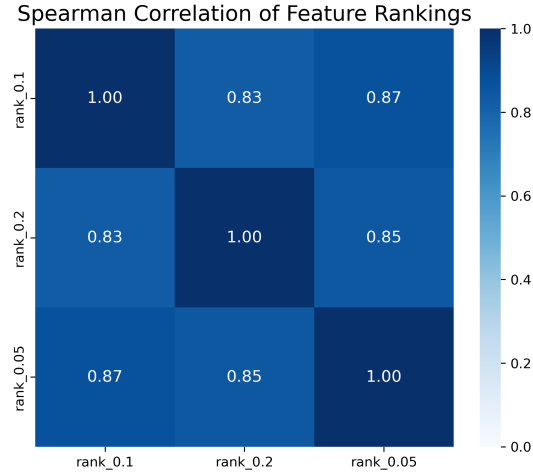


FIGURE 4.6: Spearman rank-correlation matrix for the feature-importance rankings

The next step is to test whether the contrastive model still performs well after removing the lowest-ranked features identified by the gating mechanism.

4.4.2 Contrastive after feature removal

After removing the lowest-weighted variables identified by the gating layer¹, and repeating the same experiments described in the contrastive subsection, it improved when and the model achieved better performance.

When comparing the resulting ROC and PR curves with those of the full contrastive model and the embedding Network with Random Forest classifier, the improvement is clear: after removing the least informative variables, both ROC-AUC and PR-AUC increased across all test sizes (Figures 4.7 and 4.8). Moreover, as the test size increases, the performance decreases more slowly than in the non contrastive model, showing that it's more robust when less training data are available.

These results answer the second research question: yes, a feature gating mechanism can be incorporated to reliably identify the most informative variables.

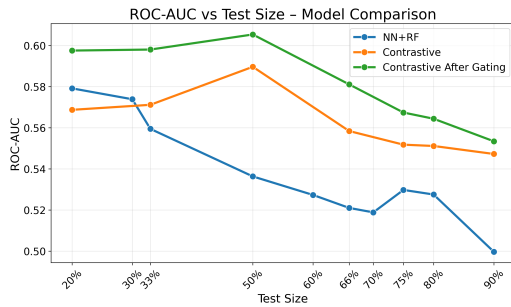


FIGURE 4.7: ROC-AUC comparison

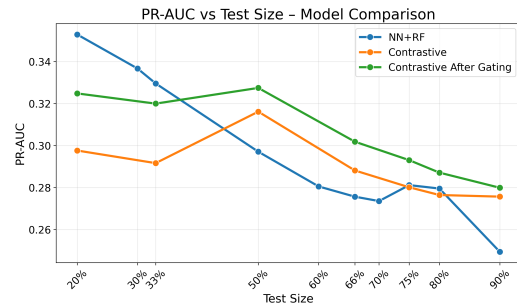


FIGURE 4.8: PR-AUC comparison

¹Specifically, we removed the nine least important features, which correspond to the lowest-weighted variables (the ones clustered near the diagonal in Figure 4.5)

Chapter 5

Conclusions

This thesis investigated whether contrastive learning can improve the prediction of venous thromboembolism (VTE) in cancer patients when working with small, high-dimensional clinical datasets. It was determined to evaluate two central research questions:

- Does contrastive learning improve the predictive performance in small, high-dimensional clinical datasets?
- Can we incorporate a feature gating mechanism to help identify the most informative variables?

Regarding the **first objective**, it was observed that, although the performance of the embedding network with a Random Forest classifier and the contrastive models decreased when the amount of training data was reduced, the contrastive approach achieved better results. Its ROC-AUC and PR-AUC curves were higher than those of the first model and showed a slower degradation when increasing the size of the test (Figures 4.3 and 4.4). This demonstrates that contrastive learning helps the encoder learn more stable and generalizable patient representations, and that is particularly well-suited for limited clinical datasets such as `oncoth2_0.2.3`.

For the **second research question**, although the gating layer did not push weights exactly to zero, it produced stable and consistent feature rankings across different regularization parameters λ_{L1} . The reliability of these rankings was validated by scatter plots and Spearman correlations across the different configurations, indicating that the mechanism successfully captures meaningful patterns of feature relevance (Figures 4.5 and 4.6). Moreover, removing the lowest-ranked variables resulted in a clear improvement in both ROC-AUC and PR-AUC metrics (Figures 4.7 and 4.8). Therefore, although the gating mechanism did not achieve strict sparsity, it fulfilled its objective as a reliable feature-ranking tool.

These findings are particularly relevant for the `ONCOTHROMB_2` project, which contains a much larger number of clinical and genomic variables while still having a relatively limited number of patients. In this scenario, contrastive learning could help improve prediction performance despite the limited sample size, while the feature-gating mechanism could help reduce the dimensionality of the data by identifying the least informative variables. Together, they offer a practical way to handle "many features, few samples", making the contrastive-gating framework a strong candidate for future `ONCOTHROMB_2` predictive models.

5.1 Future steps

Future work could try to include all the variables obtained from blood tests that were excluded in this study to avoid noise and hospital-specific batch effects. Their

inclusion would require a normalization strategy, since each hospital uses different laboratory methods, machines, units, and reference ranges. By adding these variables, more feature-importance analyses could be performed to evaluate whether they contribute meaningfully to prediction or not.

A second direction would be to incorporate genomic information, because previous prediction models (Martín et al., 2018; Muñoz et al., 2022) have shown that genetic variants improve prediction. Extending the contrastive-gating framework to these data would make the analysis more comparable to earlier studies.

Finally, what could also be done is knowledge distillation. In this approach, a larger model (the teacher) is trained using all available variables, and a smaller model (the student) learns to reproduce the teacher's behavior using only a reduced set of features. The contrastive model could act as the teacher, providing rich representations, while the student could rely on the variables identified by the gating mechanism. Distillation could enable simpler and more interpretable models that preserve predictive performance while relying on fewer inputs.

Appendix A

Preprocessing oncoth2_0.2.3

Getting started

The first stage of the project focused on understanding the ONCOTH2_0.2.3 dataset and providing an initial overview of the variables. Two main data sources are used during this process:

- `oncoth2_data`, contains the clinical information for all 432 patients. It includes demographic characteristics, medical history, cancer-related variables, laboratory measurements, and follow-up information.
- `oncoth2_traits`, serves as a metadata table that provides descriptions for the variables contained in the main dataset.

Regarding data types of `oncoth2_data`, it contained three:

Data type	Count
Numeric	49
Datetime	61
Categorical	180

TABLE A.1: Data types

After filtering and discarding the patients who didn't fulfill the study eligibility criteria nor provided informed consent, we went from 432 to 395 patients.

It's also noted that, because the study is made in Spain, the cohort is imbalanced in terms of ethnicity. However, overall sex imbalance was not something that was expected:

	Men	Women	Total
Caucasian	229	153	382 (97%)
Latin-American	3	6	9 (2%)
African	1	1	2 (0.5%)
Other	0	2	2 (0.5%)
Total	233 (59%)	162 (41%)	395

TABLE A.2: Population imbalance

When deciding whether keeping all patients and discarding blood variables, or vice versa, the possible population counts were considered (see Table A.3).

At the end it was decided that it was preferable to keep all the patients, given that there were so few, and discard the blood variables, which, even if they were taken before a VTE episode, they could still already reflect physiological alterations caused by the developing thrombosis, making them unreliable as predictive

Relation	Patients
blood test before VTE	379
same day	2
blood test after VTE	14

TABLE A.3: Timing relationship between date_blood_test and date_VTE

features. Moreover, there exists substantial variability across hospitals in how these blood measurements are obtained (differences in methods, units, ranges, etc), which introduces a batch effect that would require careful normalization to ensure comparability.

The product of these decisions left a final VTE count of

	No VTE	VTE	Missing	Total
Man	179	50	4	233 (59%)
Woman	117	45	0	162 (41%)
Total	296 (75%)	95 (24%)	4 (1%)	395

TABLE A.4: VTE and sex frequency table

Overall, this notebook provided a first characterization of the dataset and established the first preprocessing decisions that guided the remainder of the study.

Feature Selection and Data Cleaning

A very important part of the preprocessing was the removal of many variables and features. Starting with 290 variables for only 395 patients would have introduced noise and redundancy. Thanks to Bárbara Lobato (an expert familiar with both the dataset and its medical context), the total of features decreased from 290 to 59.

Still, there were some columns that were redundant and gave the same information twice. These were rs11696364, rs169713, rs4524, rs6003, rs2227631, rs268, rs5110, rs6025, rs2232698 and their corresponding n_risk_alleles. The latter set was numerical, so they were the ones kept for modelling.

The variable tumor_type_risk gave less information than primary_tumor, so the first was dropped. Furthermore, sex and menopausal_status columns gave similar information, but given that % of women were in their post-menopausal, and it didn't have any direct relation with VTE, it was considered that it wouldn't give much information and could even add more noise.

Menopausal status	VTE Yes		VTE No	
	Count	Freq (%)	Count	Freq (%)
Post-menopausal	98	86%	42	93%
Pre-menopausal	12	11%	2	4%
Peri-menopausal	4	3%	1	2%

TABLE A.5: Menopausal status distribution by VTE outcome

To assess the informativeness of the remaining features, the **variability** of each column was examined. The goal was to identify variables that contribute little or no

information because they have a very low variance (in numeric features) or very low entropy (in categorical features):

- Variance of Numeric Variables: we compute the statistical variance, defined as:

$$Var(X) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$
- Entropy of Categorical Variables: variance is not meaningful. Instead, we measure Shannon entropy, which quantifies how "spread out" the categories are:

$$H = - \sum p_i \cdot \log(p_i) \text{ for } p_i = \mathbb{P}(x_i).$$

Variance	Variable	Description
0.03	rs2232698_n_risk_alleles	# risk alleles in SNP rs2232698
0.03	rs6025_n_risk_alleles	# risk alleles in SNP rs6025
0.03	rs268_n_risk_alleles	# risk alleles in SNP rs268
0.16	rs6003_n_risk_alleles	# risk alleles in SNP rs6003
0.16	rs5110_n_risk_alleles	# risk alleles in SNP rs5110
0.19	rs11696364_n_risk_alleles	# risk alleles in SNP rs11696364
0.34	rs5985_n_risk_alleles	# risk alleles in SNP rs5985
0.38	performance_status	ECOG performance status at beginning of chemotherapy
0.39	rs169713_n_risk_alleles	# risk alleles in SNP rs169713
0.39	rs4524_n_risk_alleles	# risk alleles in SNP rs4524
0.49	rs2227631_n_risk_alleles	# risk alleles in SNP rs2227631
0.53	grade_histological_differentiation	Grade of tumor histological differentiation (well, moderately or poorly differentiated)

TABLE A.6: Variables with lowest variance in the dataset

For the numerical variables, it will be seen that the risk_alleles features with less variance increase model prediction.

The results didn't give out much information for the categorical variables, so a **frequency-based analysis** was performed to explore how strongly each one was associated with VTE. For each feature, the distribution of its categories was compared between patients with and without VTE. Using a simple score (the sum of absolute differences between category proportions across the two groups), it was easier to interpret whether a feature had potentially higher predictive relevance.

- For example, with the menopausal_status variable, the score would be (using the data from Table A.5):

$$\sum_{\text{categories}} |\mathbb{P}(\text{cat} | VTE = 1) - \mathbb{P}(\text{cat} | VTE = 0)| \sim |86 - 93| + |11 - 4| + |3 - 2| \sim 15$$

The 5 features with more score were (ordered descending) patient_status, vte_concomitant_cancer_dx, primary_tumor, hospital, performance_status. Some of them make sense: for example vte_concomitant_cancer_dx is actually a variable that is directly correlated with VTE, because it's only informed after the VTE episode has happened (in fact, this variable will be dropped for modelling).

A mixed-type **correlation matrix** was built to examine dependencies between variables of different types (numeric and categorical). It corresponds to Figure A.1.

Based on this matrix and previous discovered information, some features were removed:

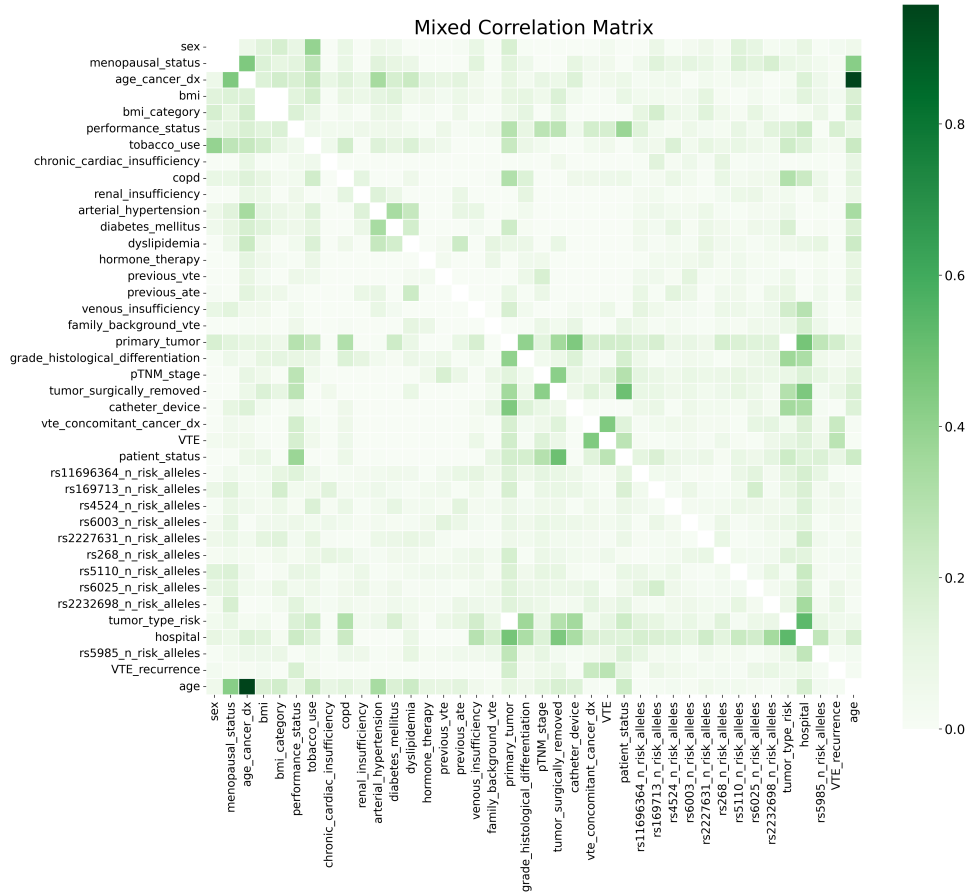


FIGURE A.1: Initial features correlation matrix

- hospital, menopausal_status, bmi_category (we keep the numerical and more informative bmi), tumor_type_risk (we keep primary_tumor), age
- pTNM_stage was labeled as categorical, but for numerical purposes, it was converted into numbers: I $\rightarrow$ 1, II $\rightarrow$ 2, III $\rightarrow$ 3, IV $\rightarrow$ 4.
- Every Y/N variable was turned into 1/0
- patient_status $\in \{\text{'Alive/Dead with/without tumour disease'}\}$ was also discarded due to its limited relevance for VTE prediction.
- Applying one-hot-encoding to tobacco_use and primary_tumor

The resulting cleaned dataset only contains 40 selected and transformed features, ready for the imputation of their missing values. Moreover, the mixed-type **correlation matrix** at the end of this preprocessing steps is Figure A.2.

Final preprocessing pipeline

The goal here is to fill in all the missing values so that the resulting dataframe is ready to become the input to all the prediction models tried in this project. This will be done by looking at the previous correlation matrix and identifying possible correlations between different variables.

The first step of the final preprocessing stage was to drop patients with missing VTE (since it is the target variable, these cases can't be meaningfully imputed).

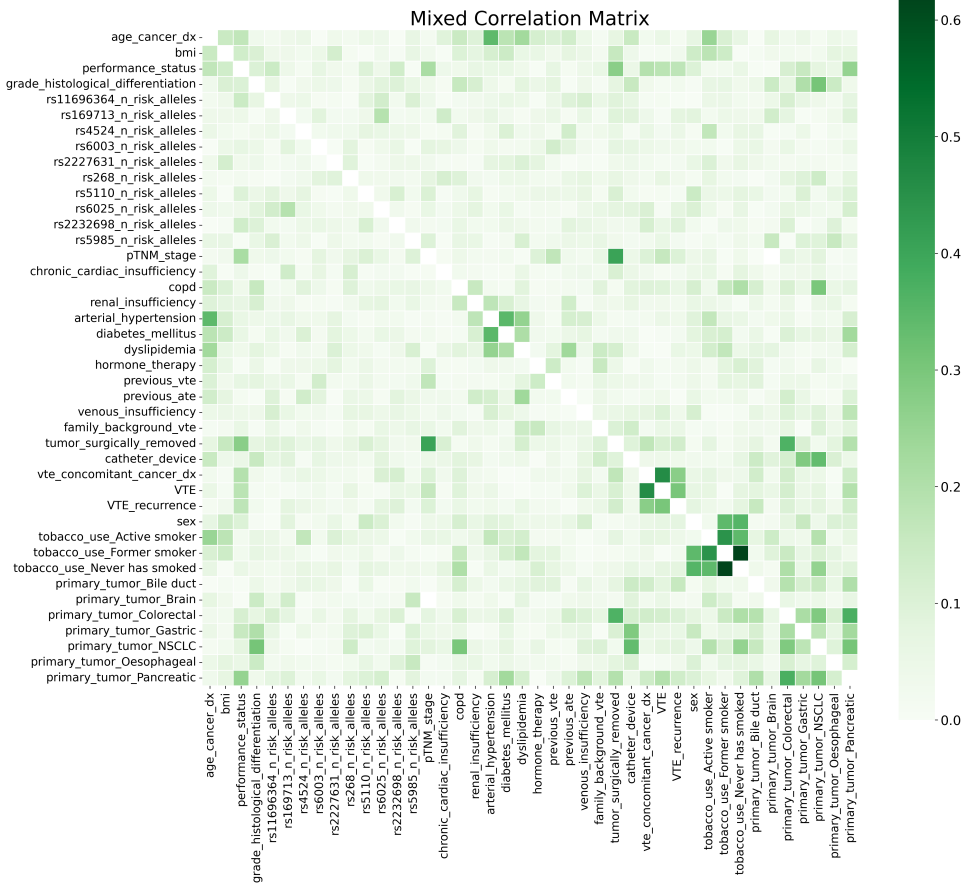


FIGURE A.2: Final features correlation matrix

Moreover, VTE_recurrence and VTE_concomitant_cancer_dx were also discarded for being defined only for VTE-positive patients.

Feature	Description
VTE_recurrence	Whether patient suffered VTE recurrence along 12 months of follow-up
VTE_concomitant_cancer_dx	Was VTE diagnosed at the same time or following to cancer diagnosis?

TABLE A.7: Dropped features for being post-VTE informative

Because the patient with the highest amount of missing data has only about 30% of variables empty, it's decided to not drop more rows, in order to keep as many patients as possible.

In order to fill in the NaN values:

- For variables showing no substantial correlation with others, missing values were filled using a function that sampled replacements according to the empirical probability distribution of the observed values.
- For pTNM_stage and tumor_surgically_removed, missing values were imputed by sampling from the observed joint and conditional probability distributions of the two variables, ensuring that the completed pairs remained clinically coherent.

- For `primary_tumor` (one-hot encoded) and `catheter_device`, missing values were imputed jointly by sampling them from their empirical conditional and joint distributions, ensuring that the built one-hot-encoded `primary_tumor` vector remained valid.

These correlation-aware procedures make sure that the new dataset maintains the empirical relationships present in the original data, avoiding unrealistic combinations of values.

After completing all imputations, the final dataset contains 391 patients, of whom 95 experienced VTE and 296 did not; the cohort includes 162 women and 229 men.

	No VTE	VTE	Total
Man	179	50	229 (59%)
Woman	117	45	162 (41%)
Total	296 (76%)	95 (24%)	391

TABLE A.8: Final VTE and sex frequency table

Appendix B

Supplementary Experimental Results

For all the experiments mentioned in this report, each setup was run across eight to ten different random states to ensure more reliable and robust results. In most cases, multiple runs were performed per random state, providing additional stability in the reported metrics.

Embedding Network with Random Forest Classifier

For the embedding Network with Random Forest classifier, initial experiments were carried out to identify suitable values for the batch size and weight decay. These tests were performed with the test size fixed at 30%, running five executions per random state. The configuration that provided the best performance was a batch size of 16 and a weight decay of 10^{-4} (see Figure B.1).

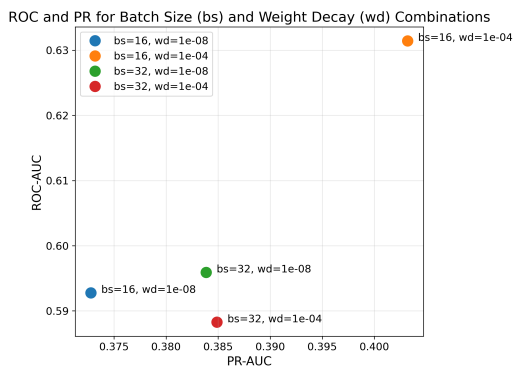


FIGURE B.1: Batch Size and Weight Decay comparison

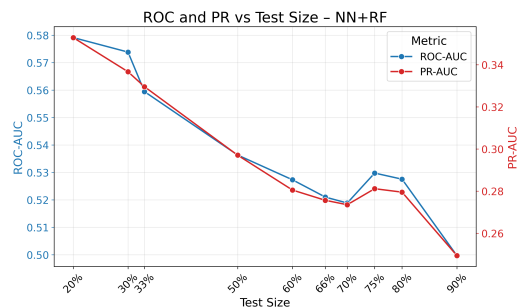


FIGURE B.2: PR-AUC and PR-AUC visualization

After fixing those parameters (along with the previous fixed learning rate = 10^{-4}), three experiments were run for each combination of random state and test size. There's also a comparison between the averaged ROC and PR curves, which follow the same overall trend, confirming that the results are stable and make sense (see Figure B.2).

Contrastive learning

For this model, only the margin parameter was tuned (fixing the test size at 20%), and it quickly became evident that a value of 0.3 provided substantially better predictive performance than the alternatives (see Figure B.3).

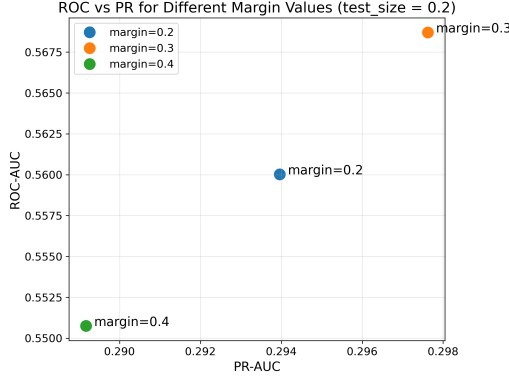


FIGURE B.3: Margin performance comparison

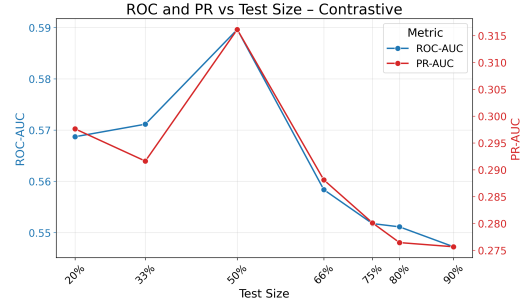


FIGURE B.4: PR-AUC and PR-AUC visualization

After fixing the parameter (along with the previous fixed batch size = 32, learning rate = 5×10^{-4} and weight decay = 10^{-8}), three experiments were run for each margin value. There's also a comparison between the averaged ROC and PR curves, which follow the same overall trend, confirming that the results are stable and make sense (see Figure B.2).

Feature gating

When adding a gating layer to the contrastive model, the training dynamics changed, so initial experiments were carried out to identify suitable values for the λ_{L1} regularization strength and the learning rate. These tests were performed with the test size fixed at 25%, running five executions per random state. The configuration that provided the best performance was a batch size of 16 and a weight decay of 10^{-4} (see Figure B.5).

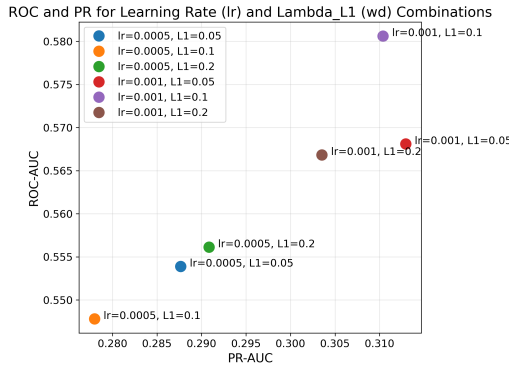


FIGURE B.5: λ_{L1} and Learning Rate comparison

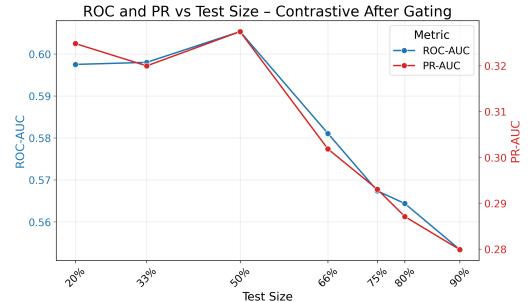


FIGURE B.6: PR-AUC and PR-AUC visualization

After fixing the remaining parameters (together with the previously selected values of batch size = 32, weight decay = 10^{-8} , and margin = 0.3), we ran experiments across ten different random states and for $\lambda_{L1} \in \{0.1, 0.2, 0.05\}$ ¹. A summary of the final feature weights obtained for the three values of λ_{L1} can be found in Table B.1.

¹Larger values of λ_{L1} apply stronger pressure toward zero, pushing more gating weights down; smaller values apply weaker pressure, resulting in fewer weights being shrunk.

$\lambda_{L1} = 0.1$		$\lambda_{L1} = 0.2$		$\lambda_{L1} = 0.05$	
Variable	Weight	Variable	Weight	Variable	Weight
sex	0.084	sex	0.035	sex	0.193
pTNM_stage	0.076	pTNM_stage	0.030	performance_status	0.189
performance_status	0.076	bmi	0.030	primary_tumor_Colorectal	0.185
bmi	0.073	grade_histological_differentiation	0.029	pTNM_stage	0.185
primary_tumor_Colorectal	0.073	tobacco_use_Active smoker	0.028	grade_histological_differentiation	0.178
tobacco_use_Former smoker	0.071	rs2227631_n_risk_alleles	0.027	primary_tumor_Pancreatic	0.169
arterial_hypertension	0.071	previous_ate	0.027	bmi	0.167
rs169713_n_risk_alleles	0.070	rs169713_n_risk_alleles	0.027	rs2227631_n_risk_alleles	0.166
tobacco_use_Active smoker	0.070	performance_status	0.027	catheter_device	0.166
tobacco_use_Never has smoked	0.069	tumor_surgically_removed	0.027	tobacco_use_Active smoker	0.164
catheter_device	0.069	primary_tumor_Bile duct	0.027	rs169713_n_risk_alleles	0.164
grade_histological_differentiation	0.069	diabetes_mellitus	0.027	tobacco_use_Never has smoked	0.164
tumor_surgically_removed	0.069	tobacco_use_Never has smoked	0.026	tumor_surgically_removed	0.164
rs2227631_n_risk_alleles	0.068	rs4524_n_risk_alleles	0.026	previous_ate	0.164
age_cancer_dx	0.068	primary_tumor_Colorectal	0.026	primary_tumor_Gastric	0.162
primary_tumor_Pancreatic	0.068	arterial_hypertension	0.026	primary_tumor_NSCLC	0.160
primary_tumor_Gastric	0.067	age_cancer_dx	0.026	family_background_vte	0.159
rs5985_n_risk_alleles	0.067	family_background_vte	0.026	diabetes_mellitus	0.158
primary_tumor_Bile duct	0.066	rs5110_n_risk_alleles	0.026	age_cancer_dx	0.158
diabetes_mellitus	0.065	primary_tumor_Pancreatic	0.025	arterial_hypertension	0.157
rs4524_n_risk_alleles	0.065	tobacco_use_Former smoker	0.025	rs5110_n_risk_alleles	0.157
venous_insufficiency	0.065	copd	0.025	primary_tumor_Bile duct	0.157
copd	0.064	catheter_device	0.025	venous_insufficiency	0.156
previous_ate	0.064	venous_insufficiency	0.024	tobacco_use_Former smoker	0.154
family_background_vte	0.064	primary_tumor_Oesophageal	0.023	copd	0.154
dyslipidemia	0.061	primary_tumor_NSCLC	0.023	rs5985_n_risk_alleles	0.154
primary_tumor_NSCLC	0.061	rs5985_n_risk_alleles	0.023	rs6003_n_risk_alleles	0.150
rs5110_n_risk_alleles	0.061	rs6003_n_risk_alleles	0.023	dyslipidemia	0.150
rs6003_n_risk_alleles	0.060	dyslipidemia	0.022	primary_tumor_Oesophageal	0.146
primary_tumor_Oesophageal	0.059	primary_tumor_Gastric	0.022	rs4524_n_risk_alleles	0.146
rs11696364_n_risk_alleles	0.054	rs11696364_n_risk_alleles	0.018	rs11696364_n_risk_alleles	0.141
rs2232698_n_risk_alleles	0.046	rs2232698_n_risk_alleles	0.018	rs6025_n_risk_alleles	0.127
renal_insufficiency	0.044	renal_insufficiency	0.016	rs2232698_n_risk_alleles	0.121
previous_vte	0.043	rs6025_n_risk_alleles	0.015	renal_insufficiency	0.117
rs6025_n_risk_alleles	0.042	rs268_n_risk_alleles	0.014	rs268_n_risk_alleles	0.103
rs268_n_risk_alleles	0.041	chronic_cardiac_insufficiency	0.014	previous_vte	0.101
chronic_cardiac_insufficiency	0.040	previous_vte	0.014	hormone_therapy	0.101
hormone_therapy	0.039	hormone_therapy	0.012	chronic_cardiac_insufficiency	0.093
primary_tumor_Brain	0.024	primary_tumor_Brain	0.005	primary_tumor_Brain	0.069

TABLE B.1: Feature weights for $\lambda_{L1} \in \{0.1, 0.2, 0.05\}$.

It is easy to see that the nine lowest-weighted variables are the same across all three settings, and the highest-ranked features also remain largely consistent. In fact, looking at Table A.6, we see that the ones with less variance actually end up with lower final importance weights.

Finally, when evaluating the model with the last 9 important features, there's also a comparison between the averaged ROC and PR curves, which follow the same overall trend, confirming that the results are stable and make sense (see Figure B.6).

B.1 Synthetic dataset

In order to have a public dataset for reproducibility while preserving patient confidentiality, a synthetic version of the original clinical dataset was generated. The complete implementation used to construct this dataset is available in the project repository; however, the core idea is as follows:

A **Gaussian Copula-based approach** was employed to model the joint distribution of the variables and to produce synthetic samples that preserve the statistical structure of the real data. I also takes into account all the data types of the database: numerical, categorical, and one-hot encoded variables. For the latter, they were temporarily reconstructed into their original categorical form and then transformed back after generating to keep the same structure.

After generating the synthetic dataset, several validation steps were performed to assess how well it reproduced the statistical properties of the real clinical data. As expected, the synthetic dataset doesn't replicate the real results exactly.

However, its purpose is to provide a statistically coherent approximation that enables full reproducibility of the analytical pipeline (see Tables B.2, B.3, B.4). All experiments using the synthetic dataset were executed using three different random states.

Test size	Real dataset		Synthetic dataset	
	ROC-AUC	PR-AUC	ROC-AUC	PR-AUC
20%	0.6248	0.3889	0.4921	0.3080
33%	0.6346	0.3791	0.4750	0.2588
50%	0.6005	0.3507	0.4851	0.2644
66%	0.5822	0.3206	0.5095	0.2856
75%	0.5802	0.3246	0.5134	0.2736
90%	0.5321	0.2772	0.4748	0.2514

TABLE B.2: Real vs Synthetic dataset on Random Forest classifier.

Test size	Real dataset		Synthetic dataset	
	ROC-AUC	PR-AUC	ROC-AUC	PR-AUC
20%	0.5791	0.3529	0.5617	0.3336
33%	0.5594	0.3296	0.5228	0.2905
50%	0.5363	0.2971	0.5294	0.3044
66%	0.5210	0.2756	0.5190	0.2998
75%	0.5297	0.2812	0.5012	0.2767
90%	0.4997	0.2494	0.4952	0.2708

TABLE B.3: Real vs Synthetic dataset on Embedding Network with Random Forest Classifier.

Test size	Real dataset		Synthetic dataset	
	ROC-AUC	PR-AUC	ROC-AUC	PR-AUC
20%	0.5687	0.2976	0.5363	0.3036
33%	0.5711	0.2916	0.5413	0.3120
50%	0.5896	0.3161	0.5406	0.2927
66%	0.5584	0.2881	0.5086	0.2789
75%	0.5518	0.2801	0.5263	0.2851
90%	0.5472	0.2757	0.5130	0.2691

TABLE B.4: Real vs Synthetic dataset on Contrastive model.

When applying the gating layer to the synthetic dataset, the nine features receiving the lowest weights were exactly the same nine variables identified as least important in the real dataset. This indicates that, despite the synthetic data not fully reproducing the predictive behavior of the real cohort, the gating mechanism still captured the same low-importance tail of the feature distribution.

However, when using the synthetic dataset, the effect of the gating layer it leads to worse results (see Figures B.7 and B.8). This means that the synthetic data do not reproduce the performance improvements observed in the real dataset when reducing the feature set based on the gating weights.

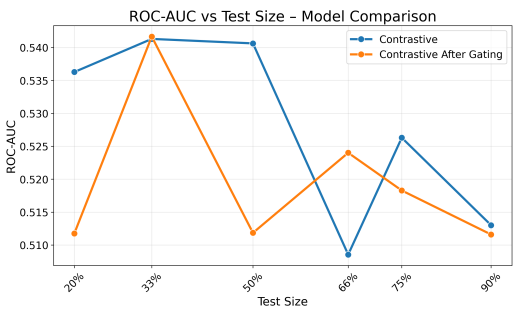


FIGURE B.7: Synthetic dataset: ROC-AUC comparison

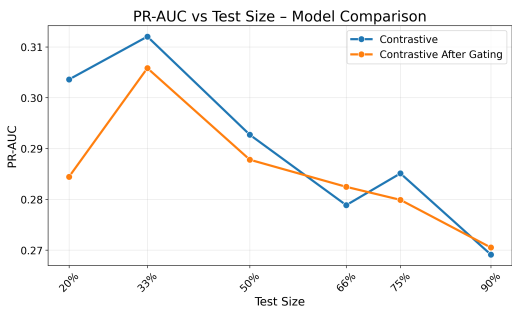


FIGURE B.8: Synthetic dataset: PR-AUC comparison

Appendix C

Source Code Repository

All the experimental results presented in this thesis were obtained using the real `oncoth2_0.2.3` dataset. However, due to privacy and data-protection regulations, this dataset can't be shared publicly.

For reproducibility, a synthetic dataset was generated to mimic the statistical structure and relationships of the original data. While the synthetic dataset captures the general behavior of the variables, its results don't perfectly align with those obtained from the real cohort. This discrepancy is expected, as synthetic data can't fully replicate the clinical dependencies present in real patients, specially if there are only 391. Still, it provides an approximation for verifying the full modelling pipeline.

The implementation code used is available at the following GitHub repository:

<https://github.com/jvc02/JVC-MasterFinalThesis.git>

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